

Interview Questions — Python - Collections, functions and Modules in Python

Total Questions: 40

1. What is a Python list, and how can it store values of different data types?

Hint: Think about how lists can hold integers, strings, floats, etc. together.

2. How would you update the value at a specific index in a list?
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3. Describe two ways to remove an element from a list in Python.

Hint: Consider both value-based and index-based removal.

4. How does a 'for' loop help you iterate through all elements in a list?
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5. What is the purpose of using 'range' with lists in a for loop?

Hint: Think about what 'range' provides compared to directly looping over the list.

6. Write a Python loop to print each element in the list ['a', 3, 5.5, 'hello'] along with its index.
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7. What does the built-in function 'round()' do when applied to a list of float numbers?

Hint: Consider if 'round' works directly on lists or if you need to use it differently.

8. Explain the difference between 'sort()' and 'sorted()' when working with lists in Python.

Hint: One modifies the list in place, the other returns a new list.

9. How would you use the 'zip()' function to combine two lists into a list of tuples?
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10. What is a tuple, and how is it different from a list?

Hint: Focus on mutability.

11. How do you create a tuple that contains both strings and numbers?
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12. Explain how slicing works for tuples. Give an example.
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13. How can you convert a tuple to a list and vice versa in Python?

Hint: There are built-in functions for both conversions.

14. How do you update the value for a specific key in a Python dictionary?

15. What is a nested dictionary? Give a simple example.

16. Given two lists: `keys = ['name', 'age'], values = ['Ravi', 21]`, write code to create a dictionary mapping keys to values using `'zip()'`.

17. Show how to create a dictionary from two lists using a loop instead of `'zip()'`.

18. How would you count the frequency of each character in a string using a dictionary?

Hint: Consider iterating over each character and updating a count.

19. Write a Python program that counts the number of words in a given sentence.

20. What is the difference between positional and keyword arguments in Python functions?

Hint: Think about how arguments are matched to parameters.

21. How can you set a default value for a function argument in Python?

22. Write a function that takes two numbers and returns their sum, but if only one number is given, the second should default to 10.

23. What is a lambda function in Python, and when would you use one?

Hint: Consider situations where you need a quick, simple function.

24. Write a lambda function that takes two arguments and returns their product.

25. How can you use a lambda function to sort a list of tuples by the second element in each tuple?

26. How do you create a custom Python module (e.g., `student.py`) and use it in another file?

Hint: Think about file creation and the `import` statement.

27. If you have a function defined in `student.py`, how do you call it from your main Python file?

28. Name two functions from the 'math' module and describe what they do.

29. How would you generate a random integer between 1 and 100 in Python?

Hint: Look into the 'random' module.

30. What is the purpose of the '__init__.py' file when creating a Python package?

Hint: Think about what makes a folder a package.

31. Explain how to structure a basic Python package with multiple modules.

32. Write a list comprehension to create a list of squares of numbers from 1 to 5.

33. How would you use a nested list comprehension to flatten a 2D list into a 1D list?

Hint: Think about looping over sublists and their elements.

34. Describe how to create a dynamic nested dictionary in Python, such as for storing student marks in multiple subjects.

Hint: Consider using dictionaries inside dictionaries.

35. Write code to add a new subject and mark for a student in an existing nested dictionary.

36. How can you use AI tools like ChatGPT or GitHub Copilot to help you write or debug Python code, and what should you always check before using their suggestions?

Hint: Think about code generation and the importance of reviewing output for correctness.

37. Imagine you need to quickly generate a function to count word frequencies in a text. How could you use an AI assistant to speed up your workflow, and how would you verify the code works as expected?

38. Integrative: How would you write a function that takes a list of sentences, counts the number of words in each sentence, and returns a dictionary mapping each sentence to its word count?

Hint: Combine knowledge of functions, lists, and dictionaries.

39. Integrative: Given a list of student names and a list of their marks, write code to create a dictionary mapping each student to their mark, and then use a function to find the student with the highest mark.

Hint: You'll need to use zip(), dictionaries, and functions together.

- 40.** Integrative: How can you use a lambda function along with the 'sorted()' function to sort a list of dictionaries representing students by their 'age' key?
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